Nonlinear Effects in Fiber-optical Communication

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An overview of fiber-optical communication is given, with an emphasize on nonlinear aspects of fiber optics. Various phase modulation effects, four wave mixing and solitons are discussed. Such nonlinear effects turn out to host potentials of improving performance in communication systems. Based on the current challenges of fiber-optic communication, methods for improving fiber-optic communication systems that exploit nonlinear properties are proposed.

I. INTRODUCTION

As the backbone of modern communication infrastructure, fiber-optic systems are crucial for the transmission of information.

Because of the small cross section of fiber core and the long distance over which optical signals travel, nonlinear effects in fibers are strong. [1] They can either disrupt or improve the performance of fiber-optic transmission systems. Turning disruption into imrpovement requires physical insight as well as engineering ingenuity.

The past few decades have seen remarkable achievements in fiber-optic communication that are aided by a better understanding of nonlinear properties of optical fiber, including ultra-long haul transmission with solitons, dispersion management via optical phase conjugation, dense wavelength-division multiplexing and space-division multiplexing through multicore fibers.

It is therefore necessary to understand and take advantage of nonlinear phenomena in fibers so that challenges are better understood and solutions become more robust.

In this paper, a historical account as well as basic concepts of fiber-optic communication will be given to facilitate an understanding of past achievements, typical techniques and open problems. Based on that, current challenges and techniques involving nonlinear phenomena will be discussed in more detail. Finally, proposals of utilizing nonlinear effects for fiber-optic communication will be made. We will also comment on the relation between the physics of nonlinear optics and the engineering aspects of fiber communication.

We do not attempt at a comprehensive analysis of fiber-optic communication *systems*, which comprise of not only fibers, but also encoders (also known as modulators), decoders and amplifiers. Instead, focus on fibers is kept throughout the paper in order for the physical perspective to emerge from various engineering technicalities. Neither will we discuss *digital* techniques in too much depth, because they are primarily implemented in the encoders and decoders instead of fibers.

II. HISTORY AND OVERVIEW

In this section we provide an overview of fiber-optic communication technogies. We start with basic concepts of communication and optic fiber to introduce core ideas as well as technical terms. More physical detail, especially those related to nonlinear effects, will be given in later sections.

A. Basic Concepts of Communication

In fiber-optic communication, a **signal** is a function of time that encodes information. If the function can take continuous values, then it corresponds to an **analog signal**; otherwise it corresponds to a **digial signal**. Any analog signal can be **quantized** into a digital signal with a certain amout of noise that is inherent to the quantization process.

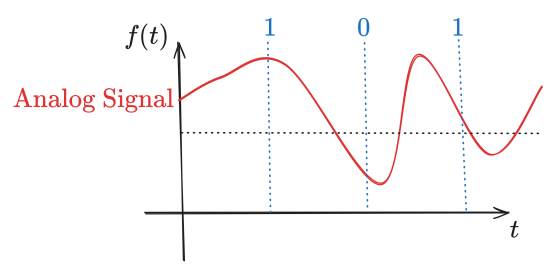

FIG. 1. Example of analog and digital signal. Red line is an analog signal; blue binary numbers are the digital signal bits after quantization. In this figure, the rightmost bit 1 contains more quantization noise than the rest two bits.

Because of the digital nature of most data that needs transmission, fiber-optical systems transmit digital signals. The **bit slot** T_B is the time duration which one bit of information occupies in the signal. The **bit rate** B is the rate at which bits are transmitted, therefore $B = 1/T_B$. A simple modulation format is to use a non-empty bit slot for encoding "1" and an empty one for "0". This is the **binary signal**. A typical quantity depicting the capability of communication systems is BL, the product of bit rate and the distance of transmission.

A **channel** is a collection of signals that are transmitted within the same physical system. In order for different signals to not disrupt each other, signals should either be delayed in time or occupy different frequencies.

Multiple channels can be transmitted within the same physical system. This is known as channel **multiplexing**, which is essentially parallel transmission of channels *without* additional physical cost.

For example, if the bit slot is longer than the pulse of one bit, then the surplus amount of time can be exploited to achieve channel multiplexing. This is known as **time-division multiplexing** (TDM). Similarly, if the physically allowed bandwidth is larger than the single-channel bandwidth, then the surplus of bandwidth can be exploited to carry multiple channels. This is known as **wavelength-division multiplexing** (WDM).

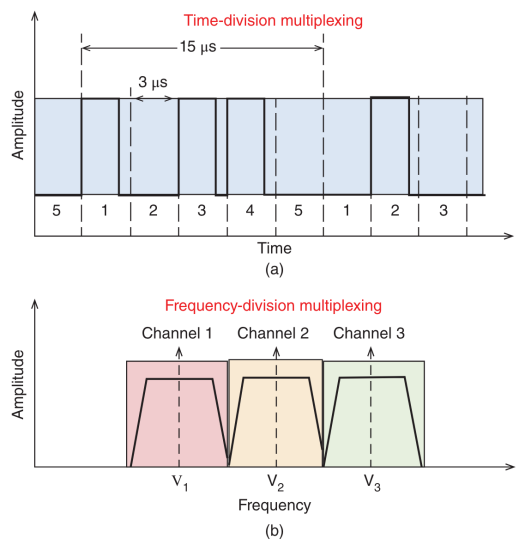

FIG. 2. Diagram for TDM and WDM. In the figure the term "frequency division multiplexing" is used intead of WDM. Reproduced from [1].

Other multiplexing techniques include **Space-division Multiplexing** (SDM) and **Coherent Detection**, which is to encode information not only in the amplitude but also in the phase of optical signals.

Multiplexing techniques can be assembled in a hierarchical manner. For example, in WDM each frequency may contain several channels that are separated via TDM or other multiplexing techniques. This fact opens up rich opportunities for multiplexing, since newer multiplexing schemes that exploit novel physical degrees of freedom can be integrated into older ones.

It should be noted that a communication channel is a well-defined object for theoretical investigation. Shannon's two coding theorems determine the upper bound of communication capacity of channels under the noiseless and noisy contexts, respectively.

Channel multiplexing is essential for improving the performance of fiber communication systems, since with engineering successes in recent decades, the capacity of one channel has come close to the Shannon limit. In fact, today's research focus in fiber-optic communication systems is precisely channel multiplexing.

B. Basic Properties of Optic Fiber

An optic fiber is a waveguide whose refractive index n decreases with the radial coordinate r. Because of the total internal reflection effect, light which are incident into optical fibers with the appropriate angle can be trapped and guided without leaking out.

Typically, a fiber consists of one core and one cladding. Multi-core fibers have been manufactured in recent years for SDM.

Fibers where $n(r)$ is a continuously decreasing function are known as **graded-index fiber**. Those where $n(r)$ experiences a sudden drop at the interface between the core and cladding are known as **step-index fiber**.

Transverse profiles of light pulses in fibers depend on the diameter and the function $n(r)$. This is a well-defined cylindrically symmetric problem of electric field with definite boundary conditions. [1, § 2.2.2] In order for the wave propagation in fiber to be better modelled, the typical approach is to manufacture the fiber so that only one transverse mode is supported. This mode is known as the **fundamental mode**, and can be approximated with a Gaussian profile. Fabricating single-mode fiber boils down to a critical value of the core diameter r_c. Given the properties of the core and cladding, cores with diameter larger than r_c will admit more than one mode.

Dispersion is the primary linear effect that disrupts signals in fiber. This effect is inherent in fibers, because typical materials that are used to fabricate fibers have nonzero dispersion. Light beams that are incident into the fiber with different angles have different optical path lengths. The dispersion caused by this effect is known as **path dispersion**. In graded-index fibers, because the spatially longer paths travel in the region where the refractive index is smaller, their optical path length would experience less increase, thus alleviating path dispersion.

The dominant dispersion effect in fiber is the Group Velocity Dispersion (GVD). The group velocity is

$$v_g = \frac{\partial \omega}{\partial k},$$

the time delay of pulses caused by difference in v_g is

$$\Delta T = \frac{\mathrm{d}T}{\mathrm{d}\omega}\Delta\omega = \Delta\omega \frac{\mathrm{d}}{\mathrm{d}\omega}\left(\frac{L}{v_g}\right) = L\Delta\omega \frac{\mathrm{d}^2 k}{\mathrm{d}\omega^2}. \quad (1)$$

If ΔT is less than one bit slot then the dispersion is tolerable for communication. This criterion is equivalent to $B\Delta T < 1$. Inserting Eqn. 1 and

$$\Delta\omega = -\frac{2\pi c^2}{\lambda}\Delta\lambda,$$

we have

$$BL|D|\Delta\lambda < 1, \quad \text{with } D := -\frac{2\pi c}{\lambda^2}\frac{\mathrm{d}^2 k}{\mathrm{d}\omega^2}. \tag{2}$$

Here, D is known as the **dispersion parameter**, which describes how severely dispersion affects communication. In Eqn. 2, we can see that for a given B, L scales as B^{-1}. This justifies using the quantity BL to characterize the communication capacity, because in order to increase B a proportional penalty in L is expected.

C. History of Fiber-optical communication

Although the capacity of fibers is upper bounded by the Shannon limit, the growth in fiber capacity has still demonstrated an exponential pattern.

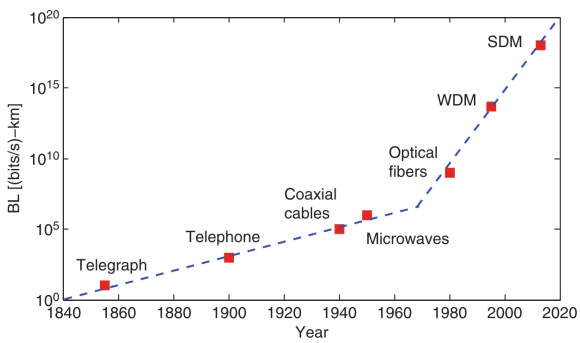

FIG. 3. Exponential growth of BL. Reproduced from [1].

According to the engineering community, there have been six generations of fiber-optic communication technology. Among them, three discoveries were key.

The first was that low-loss fiber is physically possible. Dispite the long-standing agreement that light is an excellent carrier for information, the absence of low-loss waveguide for light and coherent light sources prevented the actual implementation of optical communication. In 1970 it was realized [9] that loss in fiber can be suppressed to below 20dB/km around 1 μm wavelength. This discovery and the invention of lasers made possible the first implementation of fiber-optic communication. The major drawbacks were wide spectrum of lasers and large dispersion in fibers.

The second was that dispersion spectrum in fibers can be shifted so that the zero-dispersion wavelength matched that which was employed in communication (typically 1.55 μm). Dispersion-shifted single-mode fibers allowed the transmission of data at 4 Gbps across 100 km. [4]

The third was that signals could be amplified with optical techniques instead of digital ones, and that channels can be multiplexed by division of frequency (that is, WDM). This discovery coincided with the "internet boom" around 2000. Thanks to this innovation, transmission across over 10000 km at 10 Gbps was realized.

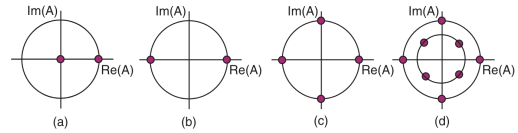

FIG. 4. Examples of coherent detection. More precise knowledge about the phase enables denser encoding. Reproduced from [1].

It is worth noting that up to this time, most techniques have centered around linear optics. However, around the same time as amplified WDM systems were developed, nonlinear effects were also being exploited to aid communication. For example, in 1993, optical phase conjugation, an intrisically nonlinear effect, was used to compensate dispersion effects. [21] Solitons [11] , the stable solutions to the nonlinear wave equation, were also used to achieve ultra-long haul (9000 km) transmission at 20 Gbps in single-mode fibers. [14] This was soon improved [13] to 1Tbps at 18,000 km via WDM. This distance is already half of earth's circumference.

Indeed, techniques after WDM have all suffered from nonlinear penalties, and nonlinear techniques that help circumvent such penalties have been widely deployed.

Today's most popular format is the so-called fifth generation communication, which is the improved version of WDM which involves larger bandwidth achieved by better fiber fabrication technologies and more bit rate enabled by coherent detection.

Around the 2010s, single-channel capacity approached the Shannon limit. [2, 3] This strengthened the belief that further improvement in fiber-optic systems could only be extracted by exploiting novel multiplexing schemes. Today's state-of-the-art fiber-optics systems are indeed advanced multiplexing systems, for example space-division systems [18] that achieve 1000 Tbps at 1000 km, and Petabit-per-second systems that combine WDM and SDM [8].

As will be articulated in the following sections, although nonlinearity can obstruct communication in many settings, techniques exploiting nonlinearity turn out to be very helpful.

III. NONLINEAR PHENOMENA IN OPTICAL FIBERS

Nonlinear material response to electromagnetic drive creates a host of phenomena that affects communication. Nonlinearity becomes appreciable under large drive intensity, which is exactly the case with optical fibers because its tiny core area. Besides, phase-modulation effects typically lead to a phase change proportional to the distance of propagation

$$\phi_{\mathrm{NL}} = \Delta k_{\mathrm{NL}} z \tag{3}$$

which is amplified by large z in practical systems.

In single-mode fibers the dominant nonlinear effect is **self-phase modulation** (SPM), while in WDM and SDM systems more nonlinear effects, including **cross-phase modulation** (XPM) and **four-wave mixing** (FWM) set in.

In this section we review the basics of nonlinear response, examine the dominant nonlinear effects in fiber optics, articulate past achievement in utilizing nonlinearity for communication, and comment on the relation between the physical aspects of nonlinear optics and the engineering challenges of fiber communication.

A. Self Phase Modulation

In fiber cores, centrosymmetry of silica glass leads to a vanishing $\chi^{(2)}$. Therefore, the dominant nonlinear optical effects is fibers are mostly $\chi^{(3)}$ effects. In the 3rd order polarization there is a term

$$\boldsymbol{P}^{(3)} = \varepsilon_0 \chi^{(3)} \boldsymbol{E}^3 = (\varepsilon_0 \chi^{(3)}/8)(3\tilde{E}_+\tilde{E}_- + c.c.)$$

which has the same frequency as $\tilde{E}$. This leads to a modified refractive index which is dependent on the optical power

$$\Delta\chi^{(1)} = \frac{3}{4}\chi^{(3)}|E|^2.$$

In fact, this can happen not only within one channel, but also between channels.

Because $k = \frac{n(\omega)\omega}{c}$, insert Eqn. 3 and $n = \sqrt{1+\chi^{(1)}}$ we see that SPM will lead to a phase shift proportional to z.

Broadening the pulse so that peak intensity lowers is a solution to this problem. Efforts in managing logal dispersion have met engineering challenges in optimizing the arrangement of local dispersion. (See Fig. III A for three different arrangements that are expected to yield the same result but turn out not to.)

B. Temporal Soliton

Temporal soliton results from the combined effort of GVD and SPM. By taking the Fourier transform on the envelope propagation equation

$$\frac{\partial \mathcal{E}}{\partial z} - \mathrm{i}\mathcal{E}\delta k = 0,$$

we obtain

$$\frac{\partial \mathcal{E}}{\partial z} + k_1\frac{\partial \mathcal{E}}{\partial t} - \mathrm{i}k_2\frac{\partial^2 \mathcal{E}}{\partial t^2} + k_3\frac{\partial^3 \mathcal{E}}{\partial t^3} - \mathrm{i}\Delta k_{\mathrm{NL}}\mathcal{E} = 0. \quad (4)$$

This is a more careful treatment of dispersion in optical fibers than the rough estimatino given in Eqn. 2. Ignoring k_3 and inserting

$$\Delta k_{\mathrm{NL}} = \frac{n_2^{(I)}\omega_0}{c}, \quad \tau \equiv t - \frac{z}{c}$$

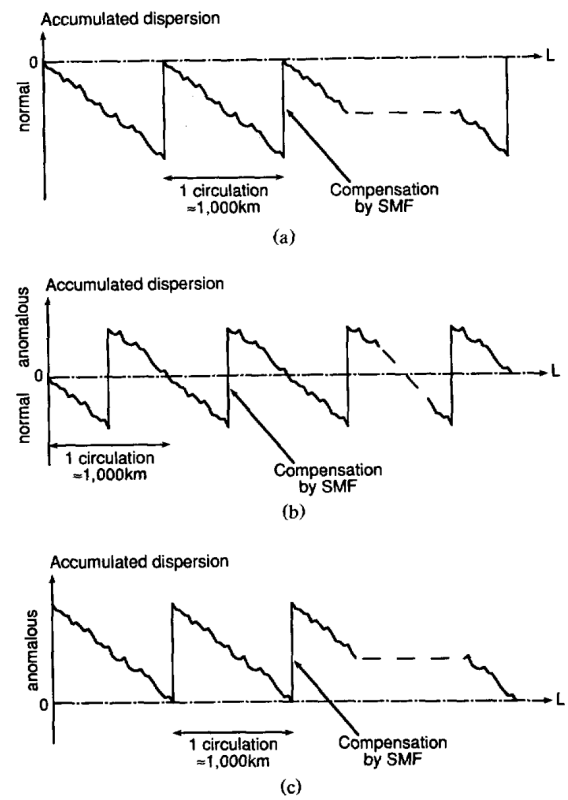

Fig. 2. A schematic diagram of the accumulated chromatic dispersion of the experiment. (a) Type I. (b) Type II. (c) Type III.

FIG. 5. Three different types of dispersion arrangement. The "anomalous" dispersion region corresponds to negative k_2 values. Reproduced from [19].

we obtain the **nonlinear Schrodinger equation** (NLSE),

$$\frac{\partial \mathcal{E}}{\partial z} + \mathrm{i}k_2\frac{\partial^2 \mathcal{E}}{\partial t^2} = \mathrm{i}\gamma|\mathcal{E}|^2\mathcal{E}. \quad (5)$$

This is the equation that governs the formation of temporal solitons.

1. *Soliton Formation*

In Eqn. 4 the term $\frac{\partial^2 \mathcal{E}}{\partial \tau^2}$ is typically negative at the pulse peak. Therefore the GVD term and nonlinear term will cancel each other only if k_2 and $n_2^{(I)}$ has opposite signs. When their effects cancel each other exactly, the pulse will neither broaden with dispersion nor broaden because of nonlinearity.

This balance between GVD and SPM gives the **soliton** solution to the NLSE, which is a family of surprisingly stable pulse. The lowest-order soliton is a sech-profile packet. [11]

In this light the dicision as for which among the three dispersion arrangements in Fig. III A becomes triv-

ial. Because the $k_2 < 0$ region effectively compensates for SPM, the nonlinear penalty will be alleviated even though there is no soliton, because the compensation is not yet enough. Therefore the optimal arrangement should be that which maximizes the propagation length in the anomalous dispersion region, which is type (c) in the figure. This aligns with the result obtained in [19].

2. *Soliton Stability*

Solitons are stable. Because this is a well-defined differential equation problem, mathematicians [16, 20] have developed relevant theories and made predictions.

There are also physical perspectives. One is that just as fibers are spatial waveguides that trap light inside, the SPM effect essentially constitutes a temporal waveguide that trappes the pulse in its most intense regions.

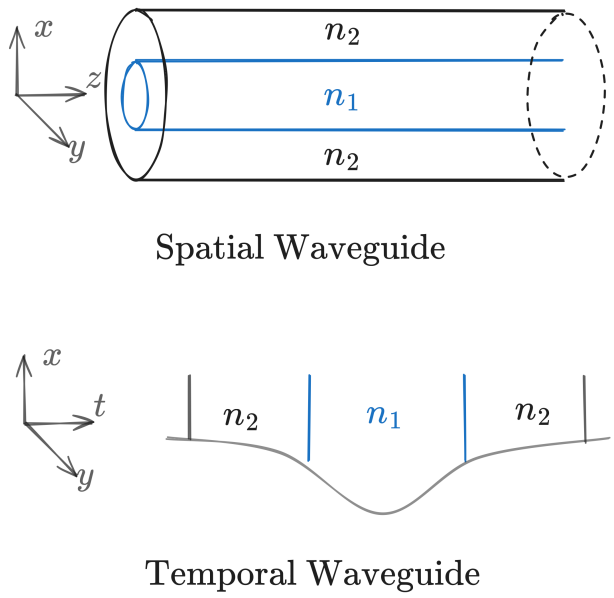

FIG. 6. Diagram that shows two kinds of waveguides. Here $n_1 > n_2$ in both waveguides.

This waveguide moves at the speed that is equal to that of pulse propagation, thus maintaining its profile throughout the fiber. Indeed, the notion that solitons are "self-trapped" pulses matches the NLSE. In the NLSE the potential energy term is

$$V = -\gamma|\mathcal{E}|^2,$$

and a positive SPM coefficient γ indicates that the potential is not a barrier but a well (see Fig. III B 2 for an intuition). However, while the Schrodinger equation describes evolution in time, the NLSE describes evolution in *space*. Directly comparing the NLSE and the Schrodinger equation will make this clear. Add to this the fact that the co-moving frame time τ instead of t appears in the equation, its exact meaning becomes opaque.

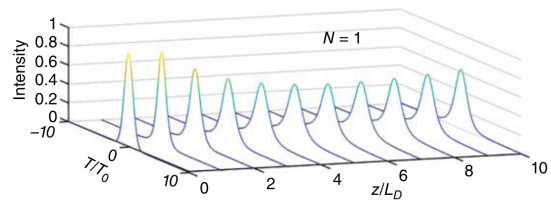

FIG. 7. Numerical result of a Gaussian profile incident into nonliner fiber. As z increases the shape approaches that of a fundamental soliton. Reproduced from [1].

Upon examining the behavior of the NLSE with only the dispersive term or only the nonlinear term, we can see that [11] both cause spreading of the "wavepacket" in the spatial coordinate. We may conclude that despite the name NLSE, intuition provided by the quantum mechanics analogy concerning this equation is limited.

One profound evidence for soliton stability is the existence of **dispersion-managed solitons**. Even though a negative $k_2 n_2$ is required for soliton formation, experiments [14] have shown that solitons can survive regions where $n_2 k_2 > 0$ as long as the *path-averaged* value of $k_2 n_2$ is appropriate.

Another evidence for soliton stability is that when non-soliton pulses are incident into fibers, the pulse shape approaches the soliton shape as it propagates. This can be viewed as the effect of a waveguide on pulses where only the supported modes survive propagation, and here the supported modes of the temporal waveguide is a series of soliton solutions.

3. *Applications of Soliton in Communication*

Solitons were initially predicted [6] and experimentally demonstrated in a commercial lab. [12] Because of their stable nature, solitons have since been candidates for fiber-optic communication, especially ultra-long haul transmission. Although GVD is balanced out by SPM in solitons, fiber loss, other nonlinear effects, and soliton collisions, can undermine soliton fiber systems' performance.

The most significant breakthrough was the implementation of dispersion-managed solitons [14]. Because in the presence of fiber loss maintaining the pulse shape of solitons is impossible, researchers gave up maintaining this shape and decided to construct the fiber system so that pulse shapes of solitons evolved periodically. This was achieved by using techniques similar to that of dispersion management (DM), where k_2 were engineered to alternate between positve and negative values as the signal propagates, resulting in the receiver a net dispersion of zero. It turned out that even after this periodic evolution the pulse shapes could be recovered quite accurately without additional digital effort. Soliton signals were transmitted across 9,000 km with DM as the only

compensation scheme.

Although solitons are not spectrally the most efficient pulses because they are sharp pulses that occupy a broad bandwidth, they can still be integrated into WDM systems. Unlike linear signals that suffer from the usual nonlinear penalties, solitons have their own unique dynamics and noise model, for example the timing jitter caused by soliton collision. In order to compensate for such effects, techniques specifically designed for soliton systems were invented. For example, Periodic-group delay method compensates for a large proportion of timing jitter, resulting in a 18,000-km soliton system operating 109 channels with WDM at a total bit rate of 1 Tbps. This distance is more than enough for the global internet, which again shows the benefit from soliton stability.

C. Four-wave Mixing

In WDM systems, channels that occupy nearby bands can interact via four-wave mixing (FWM). From a quantum optical point of view this is the annihilation of two photons and the creation of two other photons with energy conserved. This process exchanges energy between channels, thus inducing severe cross-talk noise that harms communication severely, because this is dissipative noise and can not be compensated for.

From a nonlinear optical point of view FWM can be described by

$$\frac{\mathrm{d}A_l}{\mathrm{d}z} = -\frac{\alpha}{2}A_l + d\chi^{(3)}_{\ell ijk}A_iA_jA_k^* \exp(-\mathrm{i}\Delta kz) \qquad (6)$$

with

$$\alpha = \text{ fiber loss and } \Delta k \equiv k_l + k_k - k_i - k_j.$$

For nearby channels l, i, j, k with equal frequency spacings, this effect is severe provided that Δk is small. This is the **phase matching** condition for FWM.

In this light, the cure to FWM can be DM: introducing a phase shift so that even nearby channels can not achieve phase matching during propagation, and engineer the system so that net dispersion is zero at the receiver.

D. Cross Phase Modulation

Cross phase modulation (XPM) is the inter-channel counterpart of SPM. For example, in a WDM system, between three equally spaced channels at frequencies

$$\omega_1 < \omega_2 < \omega_3$$

a polarization term

$$\tilde{P}_i^{(3)}(\omega_2) = \varepsilon_0\chi^{(3)}_{ijk\ell}(\omega_2;\omega_1,\omega_3,-\omega_2)\tilde{E}_j(\omega_1)\tilde{E}_k(\omega_3)\tilde{E}_\ell^*(-\omega_2)$$

is possible. Therefore the refractive index of channel ω_2 can be modulated by

$$\Delta\chi^{(1)} = \frac{3}{4}\chi^{(3)}_{ijk\ell}|E_jE_k|.$$

Similar to compensation against SPM, XPM can be alleviated via DM.

E. Optical Phase Conjugation

Optical phase conjugation (OPC) is an efficient technique to compensate for *single-channel* phase shift noise. Recalling Eqn. 3, if the phase of signal is reversed at the middle point $z/2$ during propagation, the total acquired phase shift would be

$$\Delta k_{\mathrm{NL}}(z + (-z))/2 = 0,$$

which is the desired result.

OPC can be achieved either through FWM or DFG; in practice it is often implemented via PPLN waveguides in a combinatino of SHG and DFG processes [7].

This method is powerful because it does not rely on any specific phase shift mechanism; as long as phase shift is proportional to propagation length, OPC can alleviate the noise significantly. For example, it can account for both SPM and GVD. Another advantage of OPC is frequency independence. In WDM systems OPC can be employed to simultaneously account for phase shift error in multiple channels. [7]

However, OPC can *not* account for inter-channel noise, such as XPM. This is because $\Delta\phi_{\mathrm{Interchannel}}$ is a function of DM arrangement and frequency difference between channels, among other parameters, which breaks down the proportional relation of $\Delta\phi$ with z.

OPC can be combined with DM to account for FWM. For example, before executing the usual OPC scheme one may run the signal through some pre-compensation fiber and after OPC the signal can be subject to a post-compensation fiber so that net accumulated dispersion at receiver is zero.

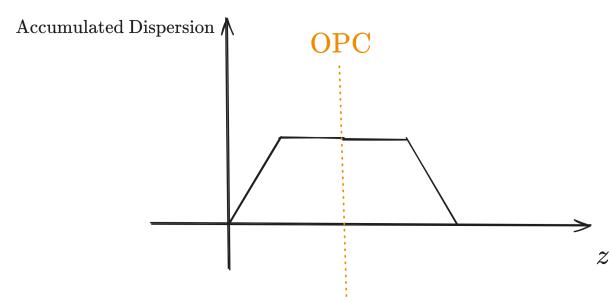

FIG. 8. Accumulated dispersion as a function propagation length.

IV. DISCUSSION AND OUTLOOK

To sum up, the various noise and compensation schemes can be summarized as shown in the below table. The table elements Y/N indicates wheter the corresponding compensation scheme is able to cancel the

noise source. Nonlinear effects constitute the primary

Noise	OPC+DM	Soliton+DM
FWM	Y	Y
SPM	Y	Y
GVD	Y	Y
XPM	N	Y

TABLE I. Summary of noise sources and compensation schemes. Note that XPM is overshadowed by collision-induced timing jitter in soliton systems.

obstacle in improving single-channel capacity as well as multiplexing efficiency. However, by exploiting nonlinear effects we have also obtained a rich collection of tools that improve fiber-optic communication systems. While soliton systems are not the focus of current research due to their low spectral efficiency and stringent modulation requirements, their stability have enabled pioneering work on ultra-long haul fiber optic transmission. More importantly, relevant scientific progress has falsified the misconception that nonlinear effects are merely incomprehensible noise that can only be circumvented through engineering and optimization. Phenomena previously deemed as engineering challenges have turned out to be physically understandable. Tapping into the rich physics of nonlinear optics has paid off well for the fiber-optic communication community.

A. Space Division Multiplexing

The current focus of fiber-optic communication is multiplexing techniques. Among the various multiplexing schemes now under development the most promising one is space division multiplexing (SDM).

As the name suggests, SDM is multiplexing channels by separating them spatially while integrating them into the same fiber. This requires the fiber to have either multiple modes or multiple cores. Non-fundamental transverse modes of fiber are more prone to noise, therefore the better choice would be to use multi-core fiber for SDM.

Quite surprisingly, multi-mode effects in SDM can compensate for SPM as well as XPM. Running multiple cores at the same time not only does not induce more nonlinear noise but also turns out to suppress it. This can be read off the following equation

$$\Delta k_i = \sum_{\ell,m,n} \chi^{(3)}_{i,\ell,m,n} E_\ell E_m E_n^* + \sum_p q_{i,p} E_p$$

with $q_{i,p}$ the linear coupling constant between different fiber cores, and $\chi^{(3)}_{i,\ell,m,n}$ the susceptibility. [15] The refractive index difference between fiber core and cladding introduces a phase shift into in-phase signals when they couple, thus producing a negative sign in $q_{i,p}$ (see the diagram for an intuition). Therefore, in SDM systems both XPM and SPM are alleviated by the inter-core coupling. In this light, manufacturers have tested multi-core fibers that are deliberately made so that coupling between cores is strong. Today's most advanced SDM systems can achieve up to 1000 Tbps at 1000 km. [18]

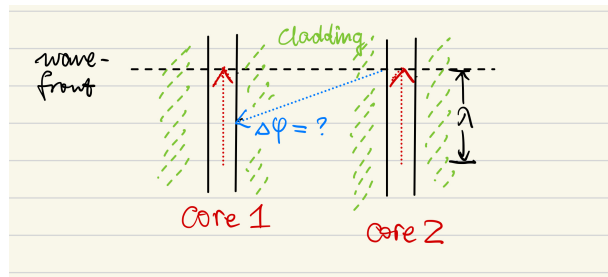

FIG. 9. Phase difference between different cores' signals lead to various effects because the transverse profile is also delayed by this phase. The blue text showing $\Delta\phi$ =? indicates that specifics depend on the physical apparatus. Here the delay is half-wavelength phase shift.

B. Nonlinear Fourier Transform

The nonlinear Fourier Transform (NFT), also known as the **backward scattering** technique, is the mathematical formalism for determining the eigenmodes of a nonlinear propagation channel. The motivation behind using this technique for communication is that by encoding information in the eigenvalues [5], more than 1 bit of information can be sent within one pulse time, and this pulse will not be subject to any nonlinear noise. The motivation is shown in the diagram below. This technique is still in its infancy, and similar to the soliton technology with wasn't competitive until challenges particular to that technology were addressed, the NFT is a promising technique that has passed numerical tests [10, 17] but awaits further development.

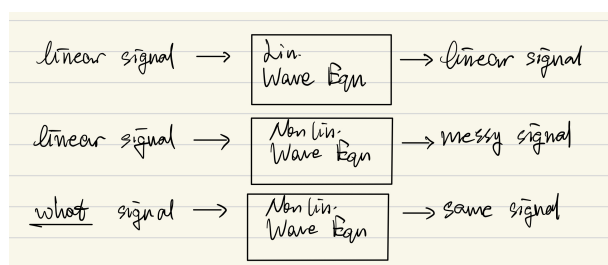

FIG. 10. A simple diagram that motivates the NFT. We already know the properties of the first two lines, and are exploring the possibility of using the third line for communication.

[1] G. P. Agrawal. *Fiber-optic communication systems*. Wiley series in microwave and optical engineering. Wiley, Hoboken, NJ, fifth edition edition, 2022. ISBN 978-1-119-73738-4 978-1-119-73737-7.

[2] Andrew D. Ellis, Jian Zhao, and David Cotter. Approaching the Non-Linear Shannon Limit. *Journal of Lightwave Technology*, 28(4):423–433, February 2010. ISSN 1558-2213. doi:10.1109/JLT.2009.2030693. URL https://ieeexplore.ieee.org/document/5210179. Conference Name: Journal of Lightwave Technology.

[3] René-Jean Essiambre, Gerhard Kramer, Peter J. Winzer, Gerard J. Foschini, and Bernhard Goebel. Capacity Limits of Optical Fiber Networks. *Journal of Lightwave Technology*, 28(4):662–701, February 2010. URL https://opg.optica.org/jlt/abstract.cfm?uri=jlt-28-4-662. Publisher: IEEE.

[4] A. Gnauck, B. Kasper, R. Linke, R. Dawson, T. Koch, T. Bridges, E. Burkhardt, R. Yen, D. Wilt, J. Campbell, K. Nelson, and L. Cohen. 4-Gbit/s transmission over 103 km of optical fiber using a novel electronic multiplexer/demultiplexer. *Journal of Lightwave Technology*, 3(5):1032–1035, October 1985. ISSN 1558-2213. doi:10.1109/JLT.1985.1074318. URL https://ieeexplore.ieee.org/abstract/document/1074318. Conference Name: Journal of Lightwave Technology.

[5] A. Hasegawa and T. Nyu. Eigenvalue communication. *Journal of Lightwave Technology*, 11(3):395–399, March 1993. ISSN 1558-2213. doi:10.1109/50.219570. URL https://ieeexplore.ieee.org/document/219570. Conference Name: Journal of Lightwave Technology.

[6] Akira Hasegawa and Frederick Tappert. Transmission of stationary nonlinear optical pulses in dispersive dielectric fibers. I. Anomalous dispersion. *Applied Physics Letters*, 23(3):142–144, August 1973. ISSN 0003-6951. doi:10.1063/1.1654836. URL https://doi.org/10.1063/1.1654836.

[7] S.L. Jansen, D. van den Borne, P.M. Krummrich, S. Spalter, G.-D. Khoe, and H. de Waardt. Long-haul DWDM transmission systems employing optical phase conjugation. *IEEE Journal of Selected Topics in Quantum Electronics*, 12(4):505–520, July 2006. ISSN 1558-4542. doi:10.1109/JSTQE.2006.876621. URL https://ieeexplore.ieee.org/abstract/document/1668091?casa_token=KzCvxIWFZi8AAAAA:0h-S9kMfSW5rBk_KzTBR3qYs_sC6IWFyuZxu15HB3jD-Q4UEoKiXboEWPOH5mINjtFPpGYfI1Q. Conference Name: IEEE Journal of Selected Topics in Quantum Electronics.

[8] A. A. Jørgensen, D. Kong, M. R. Henriksen, F. Klejs, Z. Ye, Ò B. Helgason, H. E. Hansen, H. Hu, M. Yankov, S. Forchhammer, P. Andrekson, A. Larsson, M. Karlsson, J. Schröder, Y. Sasaki, K. Aikawa, J. W. Thomsen, T. Morioka, M. Galili, V. Torres-Company, and L. K. Oxenløwe. Petabit-per-second data transmission using a chip-scale microcomb ring resonator source. *Nature Photonics*, 16(11):798–802, November 2022. ISSN 1749-4893. doi:10.1038/s41566-022-01082-z. URL https://www.nature.com/articles/s41566-022-01082-z. Publisher: Nature Publishing Group.

[9] F. P. Kapron, D. B. Keck, and R. D. Maurer. RADIATION LOSSES IN GLASS OPTICAL WAVEGUIDES. *Applied Physics Letters*, 17(10):423–425, November 1970. ISSN 0003-6951. doi:10.1063/1.1653255. URL https://doi.org/10.1063/1.1653255.

[10] Son Thai Le, Vahid Aref, and Henning Buelow. High Speed Precompensated Nonlinear Frequency-Division Multiplexed Transmissions. *Journal of Lightwave Technology*, 36(6):1296–1303, March 2018. URL https://opg.optica.org/jlt/abstract.cfm?uri=jlt-36-6-1296. Publisher: IEEE.

[11] L. F. Mollenauer and James P. Gordon. *Solitons in optical fibers: fundamentals and applications*. Elsevier/Academic Press, Amsterdam ; Boston, 2006. ISBN 978-0-12-504190-4. OCLC: ocm62290481.

[12] L. F. Mollenauer, R. H. Stolen, and J. P. Gordon. Experimental Observation of Picosecond Pulse Narrowing and Solitons in Optical Fibers. *Physical Review Letters*, 45(13):1095–1098, September 1980. doi:10.1103/PhysRevLett.45.1095. URL https://link.aps.org/doi/10.1103/PhysRevLett.45.1095. Publisher: American Physical Society.

[13] Linn F. Mollenauer, Andrew Grant, Xiang Liu, Xing Wei, Chongjin Xie, and Inuk Kang. Experimental test of dense wavelength-division multiplexing using novel, periodic-group-delay-complemented dispersion compensation and dispersion-managed solitons. *Optics Letters*, 28(21):2043, November 2003. ISSN 0146-9592, 1539-4794. doi:10.1364/OL.28.002043. URL https://opg.optica.org/abstract.cfm?URI=ol-28-21-2043.

[14] I. Morita, M. Suzuki, N. Edagawa, S. Yamamoto, H. Taga, and S. Akiba. 20-Gb/s single-channel soliton transmission over 9000 km without inline filters. *IEEE Photonics Technology Letters*, 8(11):1573–1574, November 1996. ISSN 1941-0174. doi:10.1109/68.541587. URL https://ieeexplore.ieee.org/abstract/document/541587. Conference Name: IEEE Photonics Technology Letters.

[15] Sami Mumtaz, René-Jean Essiambre, and Govind P. Agrawal. Reduction of nonlinear impairments in coupled-core multicore optical fibers. In *2012 IEEE Photonics Society Summer Topical Meeting Series*, pages 175–176, July 2012. doi:10.1109/PHOSST.2012.6280720. URL https://ieeexplore.ieee.org/abstract/document/6280720. ISSN: 2376-8614.

[16] Alan C. Newell. *Solitons in Mathematics and Physics*. Society for Industrial and Applied Mathematics, Philadelphia, Pa, first edition edition, January 1987. ISBN 978-0-89871-196-7.

[17] Jaroslaw E. Prilepsky, Stanislav A. Derevyanko, Keith J. Blow, Ildar Gabitov, and Sergei K. Turitsyn. Nonlinear Inverse Synthesis and Eigenvalue Division Multiplexing in Optical Fiber Channels. *Physical Review Letters*, 113(1):013901, July 2014. doi:10.1103/PhysRevLett.113.013901. URL https://link.aps.org/doi/10.1103/PhysRevLett.113.013901. Publisher: American Physical Society.

[18] Kunimasa Saitoh and Shoichiro Matsuo. Multicore Fiber Technology. *Journal of Lightwave Technology*, 34(1):55–66, January 2016. URL https://opg.optica.org/jlt/abstract.cfm?uri=jlt-34-1-55. Publisher: IEEE.

[19] H. Taga, S. Yamamoto, N. Edagawa, Y. Yoshida, S. Akiba, and H. Wakabayashi. Performance evaluation of the different types of fiber-chromatic-dispersion equalization for IM-DD ultralong-distance optical communication systems with Er-doped fiber amplifiers. *Journal of Lightwave Technology*, 12(9): 1616–1621, September 1994. ISSN 1558-2213. doi: 10.1109/50.320945. URL `https://ieeexplore.ieee.org/abstract/document/320945`. Conference Name: Journal of Lightwave Technology.

[20] Terence Tao. Why are solitons stable?, February 2008. URL `http://arxiv.org/abs/0802.2408`. arXiv:0802.2408 [math].

[21] S. Watanabe, T. Naito, and T. Chikama. Compensation of chromatic dispersion in a single-mode fiber by optical phase conjugation. *IEEE Photonics Technology Letters*, 5(1):92–95, January 1993. ISSN 1941-0174. doi: 10.1109/68.185071. URL `https://ieeexplore.ieee.org/abstract/document/185071`. Conference Name: IEEE Photonics Technology Letters.